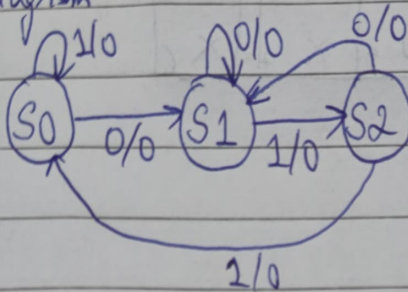


Day - 09

FSMs-Assignment

1. Draw the state diagram for a FSM, the output toggles whenever it detects '110' (LSB first (overlapping)). Consider initial value of output as 0.

Soln State diagram:



	Previous State		Input	Next State		Output (Y)
	q_1	q_0	I	D_1	D_0	Y
S0	0	0	0	0	1	0
	0	0	1	0	0	0
S1	0	1	0	0	1	0
	0	1	1	1	0	0
S2	1	0	0	0	1	0
	1	0	1	0	0	1
	X	X	X	X	X	0
	X	X	X	X	X	0

$$D_1 = q_1$$

$q_0 I$	00	01	11	10
0	0 ₀	0 ₁	1 ₃	0 ₂
1	0 ₄	0 ₅	X ₇	X ₆

$$\therefore D_1 = q_0 I$$

$$D_0 = q_1$$

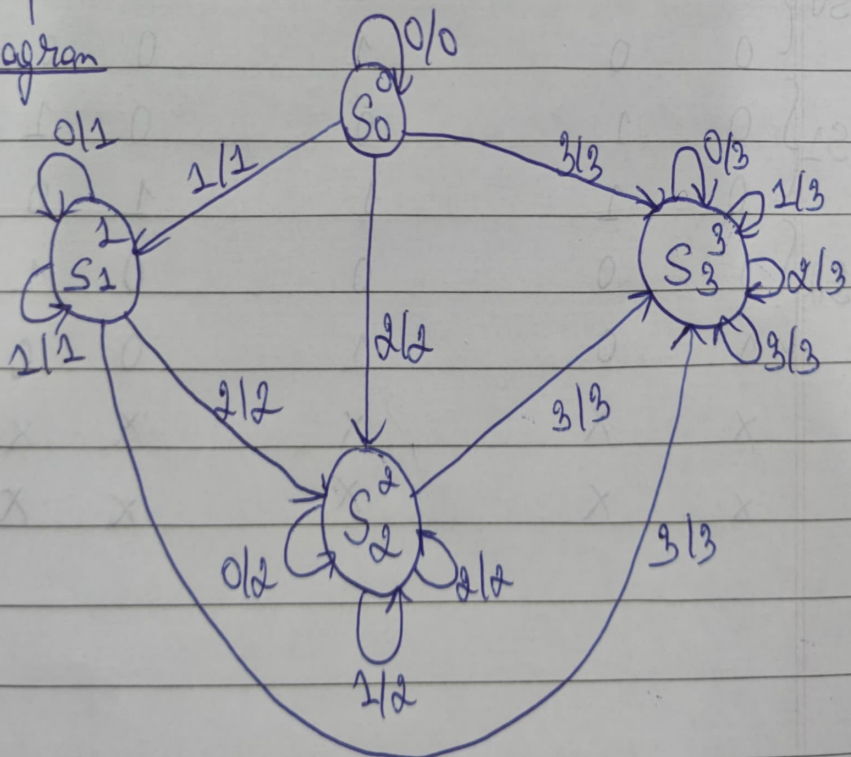
$q_0 I$	00	01	11	10
0	1 ₀	0 ₁	0 ₃	1 ₂
1	1 ₄	0 ₅	X ₇	X ₆

$$D_0 = \bar{I}$$

$$Y = q_1 \bar{q}_0 I$$

2. Design a FSM over $\{0, 1, 2, 3\}$ that will output the largest input digit read so far. e.g. input : 001032 will produce output : 001133

Solⁿ State Diagram



Previous state	Input	Next State
S_0	0	S_0
S_0	1	S_1
S_0	2	S_2
S_0	3	S_3
S_1	0	S_1
S_1	1	S_1
S_1	2	S_2
S_1	3	S_3
S_2	0	S_2
S_2	1	S_2
S_2	2	S_2
S_2	3	S_3
S_3	0	S_3
S_3	1	S_3
S_3	2	S_3
S_3	3	S_3